Slovak University of Technology in Bratislava
Faculty of informatics and information technologies

Reg. No.: FIIT-XXXX-XXXXX TODO

Timon Lumír Fillo

Correlation Power Analysis of Neural Network Weight Parameters

Bachelor's thesis

Thesis supervisor: doc. Hou Xiaolu, Ph.D.

May 2019 TODO

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Bachelor's thesis

Study programme: Information Security TODO

Study field: Computer Science TODO

Training workplace: Institute of Computer Engineering and Applied Informatics

Thesis supervisor: doc. Hou Xiaolu, Ph.D.

May 2019 TODO

Slovenská technická univerzita v Bratislave
Fakulta informatiky a informačných technológií

ZADANIE BAKALÁRSKEHO PROJEKTU

Meno študenta:
Študijný odbor: Informatika
Študijný program: Informačná bezpečnosť
Názov projektu:

Zadanie:

Práca musí obsahovať:

Anotáciu v slovenskom a anglickom jazyku
Analýzu problému
Opis riešenia
Zhodnotenie
Technickú dokumentáciu
Zoznam použitej literatúry
Elektronické médium obsahujúce vytvorený produkt spolu s dokumentáciou

Miesto vypracovania: Ústav počítačového inžinierstva a aplikovanej informatiky, FIIT STU
Vedúci projektu:

Termín odovzdania práce v zimnom semestri 10. decembra 2019
Termín odovzdania práce v letnom semestri 07. mája 2020

Bratislava 23.09.2019

Declaration of honor

I honestly declare that I prepared this thesis independently, on the basis of consultations and using the literature listed.

In Bratislava, 8.2.2020 .....

Timon Lumír Fillo

Special thanks

Anotácia

Slovenská technická univerzita v Bratislave
Fakulta informatiky a informačných technológií
Študijný program: Informačná bezpečnosť TODO

Autor: Timon Lumír Fillo
Bakalárska práca: Korelačná analýza spotreby váhových parametrov neurónových sietí
Vedúci projektu: doc. Hou Xiaolu, Ph.D.
Máj 2019 TODO

TODO: anotácia v slovenčine (150–200 slov).

Annotation

Slovak University of Technology in Bratislava
Faculty of informatics and information technologies
Degree Course: Information Security TODO

Author: Timon Lumír Fillo
Bachelor's thesis: Correlation Power Analysis of Neural Network Weight Parameters
Supervisor: doc. Hou Xiaolu, Ph.D.
May 2019 TODO

TODO: annotation in English (150–200 words).

Abstract

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List of abbreviations used

CPA	Correlation Power Analysis
DPA	Differential Power Analysis
HW	Hamming Weight
IEEE	Institute of Electrical and Electronics Engineers
MCU	Microcontroller Unit
NN	Neural Network
POI	Point of Interest
ReLU	Rectified Linear Unit
SCA	Side-Channel Analysis
TVLA	Test Vector Leakage Assessment

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Sustainability and Environmental Impact

7.1 Long-Term Maintenance

7.2 Environmental Impact

Chapter 8

Employability

Chapter 9

Teamwork, Diversity and Inclusion

Chapter 10

Scientific Part

Paper Title

***Abstract*— Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.**

Index Terms*— *side-channel analysis, template attack, neural networks

Lay Abstract

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I Introduction

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II Methods

A Subsection

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$$v = \text{float32}(w \cdot x) \tag{10.1}$$

Equation (10.1) and Table 10.1.

Table 10.1: Example table

Model	Classes	Accuracy
Coarse	18	–
Fine	24	–

B Another Subsection

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III Results

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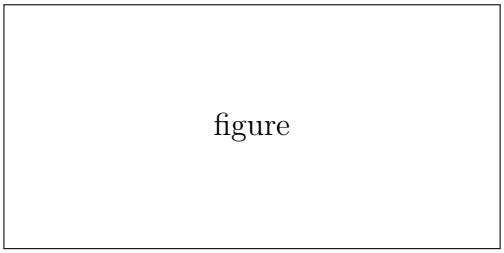


Figure 10.1: Example figure, see Fig. 10.1.

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IV Discussion

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V Supplementary Material

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References

[1] A. Author, "Example reference," *Journal*, vol. 1, pp. 1–10, 2026.

Chapter 11

Evaluation

11.1 Summary of Results

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Resumé

Appendix A

Technical Documentation

A.1 Firmware

A.2 Capture Script and Configuration

A.3 Analysis Notebooks

A.4 Data Formats

Appendix B

Work Plan

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Week	Task

B.2 Summer Semester

Week	Task

B.3 Evaluation of the Work Plan

Appendix C

Contents of the digital medium

Registration number of the thesis in the information system: FIIT-XXXX-XXXXX

TODO

Contents of the digital medium (ZIP archive):

Folder	Contents
/firmware	target firmware (network forward pass)
/capture	trace capture script and configuration
/data	
/raw	captured ChipWhisperer projects
/processed	clipped traces used by the analysis
/analysis	leakage analysis and attack notebooks
/results	exported figures and tables
/thesis	LaTeX source of this thesis
/praca-pdf	
/praca.pdf	main part of the thesis
/prilohy.pdf	text appendices

Name of the submitted archive: TODO.